

1.

In this question you must show all stages of your working.
Solutions based entirely on calculator technology are not acceptable.

- (i) (a) Use the Euclidean algorithm to determine the highest common factor h of 105 and 24

(3)

- (b) Hence determine integers a and b such that

$$105a + 24b = h$$

(3)

- (ii) Determine the remainder when 179^5 is divided by 11

(2)



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Question 1 continued

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Question 1 continued

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Question 1 continued

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(Total for Question 1 is 8 marks)



2. (i) Using a suitable algorithm and without performing any division, determine whether 13 306 617 is divisible by 9

(2)

- (ii) The group $G = \{1, 3, 7, 9, 11, 13, 17, 19\}$ has multiplication modulo 20 as its operation.

- (a) Complete the following Cayley table for G

$\times_{20}$	1	3	7	9	11	13	17	19
1	1	3	7	9	11	13	17	19
3	3		1			19	11	
7	7	1			17			13
9	9		3	1		17	13	
11	11			19	1			
13	13		11			9	1	7
17	17	11						3
19	19		13		9		3	

A copy of this table is given on page 9 if you need to rewrite your Cayley table.

(3)

- (b) State the inverse of the element 7

(1)

- (c) Determine the order of the element 13

(1)

- (d) Write down a subgroup of G of order 4

(1)



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Question 2 continued

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Question 2 continued

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Question 2 continued

Only use this grid if you need to rewrite your Cayley table.

$\times_{20}$	1	3	7	9	11	13	17	19
1	1	3	7	9	11	13	17	19
3	3		1			19	11	
7	7	1			17			13
9	9		3	1		17	13	
11	11			19	1			
13	13		11			9	1	7
17	17	11						3
19	19		13		9		3	

(Total for Question 2 is 8 marks)



- 3.** A loan of £180 000 is taken out to buy a house.

The monthly interest rate on the loan is 0.15%

The interest is added to the balance of the loan at the end of each month.

To repay the loan, £900 is repaid at the end of each month, immediately after the interest has been added.

Let B_n thousands of pounds be the balance of the loan at the end of month n after the interest has been added and the £900 repaid.

- (a) Explain, in the context of the problem, why the balance of the loan, B_n , can be modelled by the recurrence relation

$$B_n = 1.0015B_{n-1} - 0.9 \quad B_0 = 180 \quad n \in \mathbb{Z}^+ \quad (2)$$

- (b) State an assumption that must be made for this model to be valid. (1)
- (c) Solve the recurrence relation to determine a closed form for B_n (5)
- (d) Hence determine the time it will take to repay the loan. Give your answer in years and months to the nearest month. (2)

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Question 3 continued

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Question 3 continued

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Question 3 continued

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(Total for Question 3 is 10 marks)



4.

In this question you must show all stages of your working.
Solutions relying entirely on calculator technology are not acceptable.

$$\mathbf{A} = \begin{pmatrix} 6 & -1 \\ 2 & 3 \end{pmatrix}$$

Determine a matrix $\mathbf{P}$ and a diagonal matrix $\mathbf{D}$ such that

$$\mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \mathbf{D}$$

(7)



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Question 4 continued

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Question 4 continued

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Question 4 continued

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(Total for Question 4 is 7 marks)



- 5.** In an Argand diagram, the curve C with equation

$$\arg\left(\frac{z+4}{z-2i}\right) = \frac{\pi}{4}$$

represents an arc of a circle.

Given that $z = x + iy$, where x and y are real numbers,

- (a) show that this circle has equation

$$x^2 + y^2 + ax + by + c = 0$$

where a , b and c are constants to be determined.

(4)

- (b) For the curve C , determine the exact minimum value of $|z|$

(3)



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Question 5 continued

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Question 5 continued

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(Total for Question 5 is 7 marks)

TOTAL FOR FURTHER PURE MATHEMATICS 2 IS 40 MARKS

